

MediCare Lab

A reproducible symptom-classification project with transparent limits.

Software complete. Clinical capability not demonstrated. The selected random forest correctly predicts 2 of 38 held-out records (5.3%). Its accuracy is below the majority baseline. The application is explicitly an educational ML prototype, not medical diagnosis, triage or a prescribing tool.

Source records	Model study	Final evaluation
349 rows / 116 labels	195 records / 28 labels	157 train / 38 holdout
49 duplicate rows removed	4 binary symptom inputs	0 overlapping profile groups

1. Audit findings

The two archive copies are identical. Three symptom-notebook copies are identical; a fourth notebook is a distinct drug-similarity workflow. All available code, notebook cells, stored outputs, data, templates and artifact schemas were inspected. File hashes are preserved in the source inventory.

Unavailable inputs. The symptom notebook requires Training.csv and six reference CSVs that were not supplied. The drug-similarity notebook requires medicine.csv, also absent. Their stored outputs cannot reproduce those datasets or validate the old 100% accuracy claims.

Confirmed leakage and inference defects. The archive includes disease-derived risk and an outcome in the old model schema. Three scaled fields match scaling on the entire source table. The Flask app substitutes raw values for scaled values and zeros for unavailable fields, with mismatched symptom types. The model chooser still calls one model; clinical accuracy claims are unsupported.

Source-content defects. Two medication dictionaries disagree. Unverified doses and contradictory emergency directions appear in the source. The new app exposes only a traceable subset, removes dose/triage instructions and withholds Stroke recommendation fields. No medical facts were invented.

2. Foundation and limits

The complete archive CSV is the reproducible foundation. Removing 49 exact profile/label duplicates leaves 300 records. A predefined minimum of five distinct profiles per original label retains 195 records across 28 labels; 105 records across 88 labels are outside scope. No guessed label merges or Other class are introduced.

All 16 possible symptom patterns map to multiple labels in the complete dataset. There are no supplied patient IDs, dates, sites, clinical references or documented collection provenance. Full-profile grouping reduces duplicate contamination; it does not establish patient independence.

3. Reproducible evaluation

Only fever, cough, fatigue and difficulty breathing enter the fitted pipeline. Outcome, risk, pre-scaled fields and non-UI context are excluded. Binary conversion is stateless; no learned scaling or imputation is needed.

The first five-fold StratifiedGroupKFold split at seed 42 reserves the holdout. Groups hash all eight raw pre-outcome profile fields without the target. Four grouped training folds compare seven fixed classifiers; selection uses macro F1, then balanced accuracy, then predefined simplicity. The test set is evaluated after selection. The exact evaluated fit is saved, without refitting on the holdout.

Classifier	CV macro F1	CV accuracy
Random forest	0.0452	9.57%
Logistic regression	0.0291	7.66%
K nearest neighbors	0.0258	7.02%
Bernoulli Naive Bayes	0.0231	6.38%
Linear SVM	0.0215	6.40%
Decision tree	0.0142	4.44%
Dummy prior	0.0054	8.26%

4. Results and interpretation

Holdout metric	Selected random forest	Dummy prior
Accuracy	5.26%	7.89%
Macro F1	0.0298	0.0052
Balanced accuracy	2.38%	3.57%
Top-3 accuracy	13.16%	18.42%

Random forest wins training CV macro F1 (0.0452), but does not beat the dummy baseline in holdout accuracy or top-3 ranking. Its higher macro F1 does not establish medical usefulness. This negative result is retained rather than changing the selected model after inspecting the test set.

The 95% holdout accuracy interval is 0.0%-13.9%, from 2,000 resamples of 33 raw-profile groups. With only 38 test records across 28 labels, estimates are unstable. Per-class precision/recall/F1/support, error rows, CV fold scores, confusion matrices and exact split assignments are included.

Thirteen symptom patterns occur in both partitions through different raw profiles. An additional training-only stress test holds out entire symptom patterns: mean macro F1 0.0037. It is a separate robustness challenge, not a selection criterion. No calibrated clinical probability or AUROC claim is made.

Method references: [scikit-learn leakage and preprocessing guidance](#); [StratifiedGroupKFold](#). These support technical practice only.

5. Finished application

Symptom explorer. Users select from the four supported symptoms and confirm that unselected symptoms mean absent. The app displays a model label, two other ranked labels, matching-training-record ambiguity, and a prominent educational notice. It rejects empty input and removes stale results after input changes.

Condition reference. All 116 exact source labels can be browsed. Ten have supplied reference records; nine expose a limited, unverified subset behind an explicit study checkbox. Each name or excerpt points to the source field/index. Descriptions are unavailable and are labelled missing. Drug names are not treatment suggestions; no doses or substitute medicines are supplied.

Model and data. The interface exposes actual metrics, baseline performance, exclusions, class coverage and figures. It collects no names or contact details, stores no symptom history and uses no external medical API. Usage telemetry is disabled and local launch binds to localhost.

6. Testing and deliverables

31 automated tests cover data scope, split/CV isolation, feature ordering, invalid inputs, all 15 non-empty symptom combinations, model reload parity, metric recomputation, missing/corrupt model files, exact source links, missing/withheld references and real Streamlit execution. The executed notebook independently reproduces selection, scores, predictions and splits.

The live Streamlit server was additionally exercised in Chromium. Desktop and mobile layouts, prediction, reference disclosure and model navigation were checked. Actual screenshots and browser-check results are included.

Included: application and reusable Python modules; exact requirements; original CSV and source-reference JSON; saved model and checksums; executed final notebook; README; source audit; model card; full evaluation tables and figures; automated tests; reproduction scripts; this report and browser screenshots.

7. Run and next steps

Use Python 3.12 in a virtual environment. Install requirements.txt, then run **python -m streamlit run app.py**. The README includes Windows PowerShell commands that do not require environment activation. For testing and notebooks, install requirements-dev.txt. Reproduce into a separate folder with **python -m healthcare.train --output-dir reproduction**.

Before any clinical research extension: recover the missing datasets and their provenance/reuse terms; obtain richer governed symptom records with patient, site and time identifiers; define intended use; clinically review every reference entry; then perform a new, independently planned external evaluation. Do not repeatedly tune against this holdout.

No clinical deployment, medical certification, personalization, demographic fairness validation, or reliable diagnosis is claimed. Source reuse permissions remain undocumented.

Testing reference: [Streamlit AppTest documentation](#). Full technical and source audit details are in the accompanying repository.